

Kavya Yeluguri

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Professional Experience

Data Engineer – PayPal, Chennai, India

Nov 2024 – Present

- Architected end-to-end distributed pipelines on Google Dataproc processing ~25 TB of transactional data across 100M+ rows at 30-minute intervals under a 60-minute SLA with 99.99% pipeline availability.
- Optimized Spark transformations and Medallion storage layers on BigQuery using nested types, partitioning, and clustering, improving query performance across 25+ downstream analytics workflows.
- Prototyped a real-time streaming pipeline POC using Google Pub/Sub and Dataflow for event-driven analytics.
- Established an automated data quality and validation framework across 30+ pipeline stages—including null validation, unique key constraints, type enforcement, and special-character sanitization—reducing data incidents by 15%.
- Automated pipeline health monitoring and Slack-based failure alerting for Airflow-orchestrated pipelines in development environments, eliminating manual monitoring and improving operational efficiency by 25%.
- Built an AI-assisted incident triage system integrated with Jira to automatically generate diagnostic findings in ticket comments, cutting manual triage effort by half.
- Partnered with product and business stakeholders on technical execution, while mentoring junior engineers and supporting delivery across project workflows.

Senior Data Engineer – HSBC, Hyderabad, India

Nov 2021 – Nov 2024

- Delivered ETL pipelines on Google Cloud Platform, boosting data processing speed by 50% and reducing system downtime by 30%.
- Constructed Looker Studio tracking dashboards to monitor BigQuery cost metrics across 20+ user workloads, identifying resource inefficiencies to cut overall query costs by 30%.
- Developed Python ETL workflows handling multi-format data ingestion (CSV, JSON, Parquet), applying Test-Driven Development (TDD) with Pytest to cut production defects by 40%.
- Implemented complex SQL data models on Cloud SQL using stored procedures, views, and triggers to optimize data transformation across downstream analytics pipelines.

Data Engineer – HSBC, Hyderabad, India

Aug 2021 – Oct 2021

- Configured AWS Lambda-based microservices to integrate multiple backend APIs for e-commerce transaction processing.
- Managed payment gateway API integrations and integration testing, ensuring strict payload validation, data privacy compliance, and high availability for live banking transactions.

Projects

Claude-MCP Data Connector

- Created a custom Model Context Protocol (MCP) server using FastMCP and Python to expose BigQuery tools to Claude, enabling natural-language querying with schema-grounding guardrails to prevent unnecessary table scans.

Real-Time Crypto Streaming Intelligence Platform

- Designed and built real-time ingestion platform consuming live trade streams from Binance and Coinbase WebSockets via Apache Kafka.
- Engineered stateful Flink streaming pipelines for cross-exchange trade analysis, routing processing events and real-time operational metrics to Grafana.

Technical Skills

Languages: Python, SQL, Bash / Shell Scripting, Java

Distributed Processing & Streaming: Apache Spark (PySpark), Apache Flink, Apache Kafka, Apache Beam, Google Cloud Dataproc, Google Cloud Dataflow, Google Cloud Pub/Sub

Databases & Storage: Google BigQuery, Google Cloud SQL (PostgreSQL), Google Cloud Bigtable, Oracle

Cloud & Serverless Platforms: Google Cloud Platform (GCP), Amazon Web Services (AWS)

Orchestration & Tools: Apache Airflow, Pytest, Docker, Looker Studio, Grafana

Core Competencies: Distributed Systems Design, Data Modeling, ETL/ELT Pipelines, Data Quality & Governance

Education

CMR Institute of Technology

Bachelor of Technology in Computer Science (GPA: 9.1/10.0)

Hyderabad, India

June 2017 – June 2021

Honors & Awards

- Circle of Excellence | Pat on the Back